

Chaos Chess

AI Game

<https://github.com/Draquian/AIGameCGL>

This game is a new digital version with new features of the very know board game called Chess, like the original game, we keep the original pieces with their unique movements (an exception of the *Castling* and *En Passant*) and the shape of the board, we also maintain the colors of the modern chess (white and black), but in this new version we can find so many new, chaotic, and extreme differences. The first we can saw is that the board is a little bit bigger, we change the size 8x8 to 10x10, this not seams a big change, but without this we can add the other new features, the next what we need to focus is in the new pieces with some new movements and abilities and combinations, we add 4 new pieces (Assassin, Soul Eater, Priest, Chaos), and because of that, we need to redistribute some of the pieces:

Rook	Knight	Bishop	Priest	Queen	King	Chaos	Bishop	Knight	Rook
Assassin	Pawn	Pawn	Soul Eater	Pawn	Pawn	Soul Eater	Pawn	Pawn	Assassin
Assassin	Pawn	Pawn	Soul Eater	Pawn	Pawn	Soul Eater	Pawn	Pawn	Assassin
Rook	Knight	Bishop	Chaos	King	Queen	Priest	Bishop	Knight	Rook

As I said, the original pieces of the chess follows the original rules and movement:

- Rook: Straight line forwards and backwards through any square on the board that isn't occupied by another piece.
- Knight: Two squares vertically and one square horizontally, or two squares horizontally and one square vertically, jumping over other pieces.
- Bishop: In diagonal through any square on the board that isn't occupied by another piece.
- King: Only one square horizontally, vertically, or diagonally
- Queen: A combination of the rook and bishop movements.

The new pieces have some new rules and movements:

- Assassin: Like a King but with a two square range and can only *destroy* (this piece destroy, don't capture) by attacking from behind (it's considered behind when the first assassin jumps over that piece).
- Soul Eater: It's a combination of the Knight and Bishop movement, when they capture an enemy piece, they copy the ability or movement for 1 time (they can keep this using their normal movement until they capture another piece or using the movement / ability that they get).
- Priest: Like a King, but this piece can't capture other pieces and also will be able to *revive* pieces captured within the last 2 turns if it reaches an adjacent square. If the card is *destroyed* and not captured, this can't be *revived*.
- Chaos: One square in any direction (like a King) or up to two squares horizontally or vertically. This piece also has the ability to generate a card every 3 total turns (not personal turns, all turns) until you have 5 as maximum, this ability can only be used when this piece is on the board.

Each card has their unique description, ability and category; depending of the category, they have different probabilities to be generated:

- Teleportation: Common card, this card teleports 1 of your pieces of your choice to an empty square.
- Swap Positions: Common card, this card changes position between 2 of your pieces that you select.
- Freeze: Common card, this card makes 1 of the enemy pieces of your choice to be unplayable for 2 turns.
- Tornado: Common card, this card teleports any of the pieces of a 3x3 random area to a random empty position on the board.
- Trap Detector: Rare card, this card detects if there's a trap in an area of 3x3 on the spot that you select.
- Trap: Rare card, this card generates a trap in the position of 1 of your pieces that you select, this trap kills the first piece that will touch.
- Sacrifice: Rare card, this card sacrifices 1 of your pieces (this is *destroyed*) that you select to have a chance to *destroy* a similar enemy piece.
- Revenge: Rare card, this card adds the revenge mode in a piece that you select, when this piece is captured, the other piece is captured.
- Banish: Epic card, this card makes a piece of your choice invisible for the rest of the game unless a reveal card is used.
- Reveal: Epic card, this card reveals all the traps and banished pieces on the board.

- Promote: Epic card, this card lets any piece (except king's) be promoted at any moment.
- Summon: Legendary card, this card spawn 2 assassins one in each lateral and 1 pawn in the back square of the chaos piece if those squares are empty.

Last and not least, we find the AI casting the game like an epic medieval war, this is a text where you can read what is happening on the board like it was a big combat between 2 great kingdoms, for this I use 4 different prompts to send to the AI: a context prompt to keep the type of casting and some context about the situation, another prompt with negative inputs to limitate the output like thinks to don't show or the size of that (this can be improved with more playtesting, always that something that we don't want, we need to add to this prompt that), and for the last prompt we send the game actions.

To get the game actions we use the function "GetPrompt()" in the script "LLMComunication" that gets a string and use that string as prompt, for this in the different parts of the game manager we collect all the events in a string and at the end we send that string, in case that 1 separate event is happening we send a different string directly to the LLM, that type of events can be: game starts, the game end, promotions or card abilities.

The controls of the game are very simple:

- Right click: interact with the cards and their abilities.
- Left click: interact with the pieces and the board.
- Press R to reset the game when you want.
- Press any key to reset the game when this is over.
- Press H to open the documentation and get help.
- Press Escape to quit the application.

***Note:** the IA module that i was using is the "**mistral-7b-instruct-v0.2.Q4_K_M.gguf**" and the chat template is the "**llama 2**", whit the AI i was a big problem that in my usual PC takes around 1 or 2 hours to make a word, and when i change to another PC i needed to delete the AI model and download again because the Unity is saying me that the model exist but when i play the game i doesn't work and give me an error.

In the idea/design part I use the free ChatGPT (sometimes 3 and sometimes 3.5) and I intervene a little bit in some aspect like given the base idea of the game (a chess game 2.0 with more options and more interactive), chose between some path to take that AI give me, some changes in some ideas to have more variety or make some stuff more balanced (like adjust some movements or abilities, adapt the abilities and their numbers, or discard some ideas), for the position of the pieces I try to use the AI but i don't get anything good so i don't use any AI for this, I just try it a few times, but sometimes only replaces all the original pieces with the new ones or other times just ignore the command and don't use the new ones.

For the new pieces I start asking for a big list of new pieces with a small description that can match with the original game and take a few that I like and feel can make a good game, I made this a couple of times until I have 4 new pieces that I like. With this information I ask the AI to show me 5 possibilities to make those pieces unique, and they give me some ideas that I take a look at and choose the most accurate of the list (for some pieces I ask this more than 1 time). When I have all the pieces, the only one that has an external ability that I have to focus more on the design is the Chaos, because I need to design all the cards' abilities. So as I had with the new pieces I made the same with the cards, I asked for a big list (20 items) of possible cards that match with all the game we made until that moment. In the first iteration i keep some of the cards, and i asked to the AI that generate more items but keeping those I like (and i give the number of the list), I made this 3 or 4 time but the most part of those list was the same with different names (or sometimes with the same name), so I take those that I like and some name ideas (that at the end I design based on the names) and I ask that given the small description that AI made (if i change or made new i given to them) they give me better description of the cards and they abilities, and to finish this I ask to make categories for this cards based on the description that AI made and how strong they are, I give the categories (common, rare, epic, legendary) and then when I have the list of each category i made some changes based on how strong I see the cards.

Then in the code part, I start asking if they can give me the code to make the famous game Chess in C# for Unity, the start giving me code and some explanation about the code and the chess, but I saw that I need more base code that the AI was giving me, so I ask for a small part of the code, and I ask for the code to make the procedural board, so I can set the size I want, then I ask for a script to generate the pieces in the board, making the black side a mirror from the white side, they give me the code with a list of piece to fill.

Then I needed to be able to select those pieces and tiles, so I ask the code and the AI assume that the pieces was in array and the tiles was in another array, when I ask that part the AI told me that that can be fixed with the tags (for the white and black pieces) and layers (for all pieces and for the tiles), so I give the prefabs pieces and tails their corresponding layers, and by code the tag if their are black or white pieces, and then I ask for the turn selection and turn skip.

When I make sure that i can select select all the pieces in each turn and I can detect the tiles, I ask if we can starting the movement code, the first that I made was the *Pawn* and that doesn't give me any problem and I ask the code to capture the other pieces, then I ask for the code for the movement of the *King* and this was easy to, then I ask for the *Rook* movement and this movement give me some problemes because I need to check if was any piece in the path, but with some prompts the AI give me a code that works, the same for the

Bishop, this time the problem was more easy to solve, with the *Queen* was so easy, I just call the functions of the *Bishop* and *Rook*, then I ask for the *Knight* (I not was sure if it gonna be easy or the most difficult) but it work with the first prompt. Then I completed the original game with the ability *Promote* for the *Pawns*, I asked for this code and the AI provided me with a code and told me that I need to set the menu in Unity and add to the buttons of that menu the function of *Promotion*.

At this point we has almost all the original game, so I start to ask to focus more in the new pieces, at first I ask for the *Assassin* movement code, that it was the *King* but at 1 or 2 squares distance, that was easy (I was worried about the unique ability of killing after jumping the piece), after that I remember to the AI the ability of this piece and they give me a new script and told me to put this new script to the *Assassin* piece and make small remake to the script of the movement to add the killing enemy pieces part.

This was the first of the new pieces, but now I need to focus on the following pieces and I decided to make the *Soul Eater*, I ask if the remember this piece (the give me again all the information we had about this) and ask for the movement code, the AI gave me a code that combine the *Knight* and *Bishop* movement and the *Knight* part works very well, but in the *Bishop* part I need to iterate some more times. Then I ask for the ability code, like the previous piece, the AI gives me a code for a new script to attach to the piece that looks when the piece captures an enemy piece and takes the name of that to copy the movement (and add another script of ability if it's necessary), I need to modify the part of taking the names of the captured pieces (but not so much). At this point, some code that they gave me but like old code repeated was very different that just ignored and adapted the new code to work with my current code.

The next piece that I focused on was the *Priest*, I didn't ask for the code, I just copied the function of the *King* movement except for the part that if the selected square has some piece (of any team) don't accept that movement. When already has the movement working I ask to ChatGPT how to make the ability of this piece, they give me a code that add 2 list of the class *CapturedPiece* to the original code and the code for this class that only has the piece, the turns that they can be revived and the position. Then I ask how can add this to my code and they give me a example, when I test the new code I can find some problemes, like if in that square are some piece (at this moment the piece revive but then both pieces die if are enemy pieces or both are in the same square if they are in the same team), so I ask for a solution to those problemes and AI gave me a similar but different code for the *Priest*, after some testing and adapting the code with the help of AI, it works.

At this point only 1 piece remains to make the code, the *Chaos*. As I made before, I asked for the information of this piece that we conceived, some of this information was different from before, so I need to ratify the missing or wrong information. When the information was correct I ask for the movement code, this code was part similar to the *King* and some other part similar to a limited *Rook* movement. The new base code that the AI gave me was so different that I had some problems with the implementation, but I added very well. Then I check if the movement works (not only this, all with the new code) and I find some problems with the *Rook* and *Queen* that I ask the AI for a different approach so I can solve them, and I do. When all problems that I can find were solved, I ask for the ability code for this piece, as they do before the AI give me a code in a new script, but I need to edit almost all, at this

point the code that the AI was given to me are so different to the original that I was harder to adapt to my code that to made it myself, so I the base of this script is made by me (in this script we find as well some of the abilities, those were made using mainly the AI).

After checking the ability of the *Chaos* piece I start (like always) to ask to the AI for the cards that we decided before, when I had all the information about the cards I ask for the first 1 and followed by the next until we finish, in the process I decided that some of those cards are very unbalanced or doesn't feed well in the game, so I change it. (Approximately half of the card code is made by AI without adaptation, the other part is that I was faster than AI or I needed to change some parts of the AI code and some other parts was code used before that I reused).

When I finish the main game and functionalities, I testes, but something was feeling weird (apart of not having art), the cards doesn't had description, so I made a fast script with the help of ChatGPT to make a small description (the cards descriptions was made by the AI) that shows when you touch one of the cards, then I tested again, but it feels weird, the player doesn't see the change of turns, so I made without the AI help a code to change the cards size and colors, and to change the camera to the POV of each player, and in the process I add some settings to the scene.

Then for the art I ask for descriptions of how it looks like the pieces and the cards to the ChatGPT after I don't achieve it by myself.

Find my conversation with ChatGPT about this game and the help provided in the following link:

- <https://chatgpt.com/share/bc1a4763-c9c0-441d-bb60-03d16b3d158d>

For the art part I try so many AI, mostly all for the UI sprites but I used one for the 3D pieces

- 3D Pieces:
 - <https://charmed.ai/> this AI generates a draw based on a prompt and based on that draw that you chose they give you a 3D model (not very smooth)

***Note:** the kings doesn't have a 3D model to make sure that you can select (in the assets ara a model for the kings) because with the given models by this AI the mesh always have some problems and cause me some problems with the selection, this are not the only models that cause me problems but it's the most trouble mesh piece.

- UI Sprites:
 - <https://www.canva.com/ai-art-generator/>
 - <https://openai.com/index/dall-e-2/>
 - <https://www.adobe.com/products/firefly/features/ai-art-generator.html>
 - <https://www.artificialstudio.ai/tools/create-game-items>
 - <https://www.midjourney.com/home>
 - <https://www.fotor.com/images/create>
 - <https://www.artguru.ai/es/>
 - https://openart.ai/home?gad_source=1&gclid=CjwKCAjwjeuyBhBuEiwAJ3vuofYVI823PRtiZHI3GSfVd6vGsu6XE1qabgKviKxLxW3GWS57r_qNZhoCPqoQAvD_BwE
 - <https://deepai.org/machine-learning-model/image-editor>
<https://deepai.org/machine-learning-model/text2img>
 - <https://gemini.google.com/app?hl=es-ES>
 - <https://designer.microsoft.com/home>

After trying all of this AI I mostly don't like the results that the AI's gave (or they make me pay) me except for the AI designer of microsoft and the deepai, but the results of this last one are not very consistent to maintain the same line, so at the end I only use the art generated by designer (microsoft AI) using the prompts that ChatGPT give me, and then I need to edit to remove the background (in the project are the original and edited sprites).

The main problem that I have in this part was the limited prompts that I can do for a day.